Bachelor of Computer Applications Semester – V Business Analytics
Experiment No.: 4
Title: Data Cleaning and Preparation Using POWER BI

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Github:
https://github.com/abhi7abhi/BA

1. Introduction

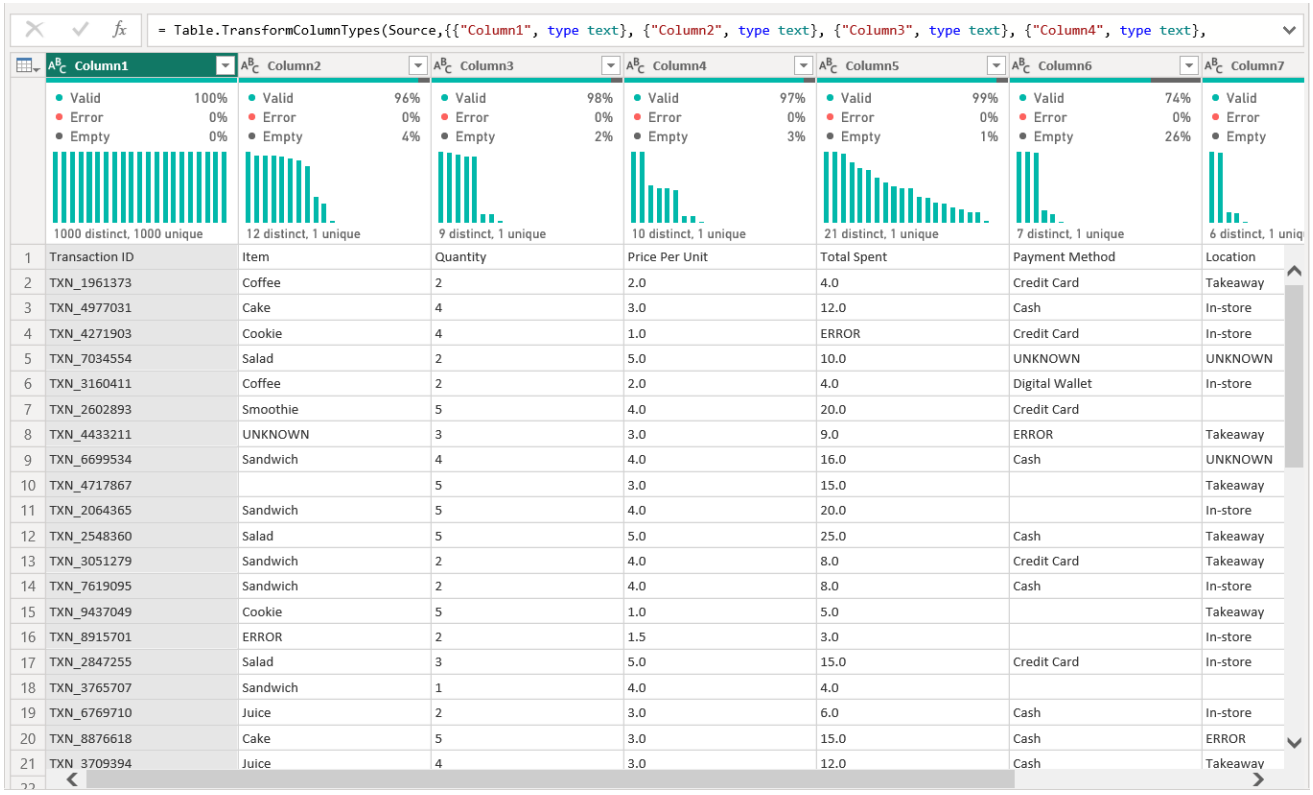
This report documents the cleaning and validation of a café sales dataset in Microsoft Power BI using Power Query. The raw dataset contained incomplete and inconsistent entries, including ERROR and UNKNOWN values, as well as fields that required appropriate numeric and date data types. The objective was to produce a reliable dataset for analysis while maintaining a clear audit trail of the transformation steps.

2. Objectives

- Identify missing and invalid records before analysis.
- Standardize data types and remove conversion errors.
- Create a calculated total using Quantity × Price Per Unit.
- Validate calculated values against the source Total Spent.
- Prepare a clean table suitable for Power BI reporting and visualization.

3. Initial Data Condition

The initial Power Query view showed raw values such as ERROR and UNKNOWN. The data-quality profile also showed empty values in several columns. These issues had to be resolved before building the analytical model.



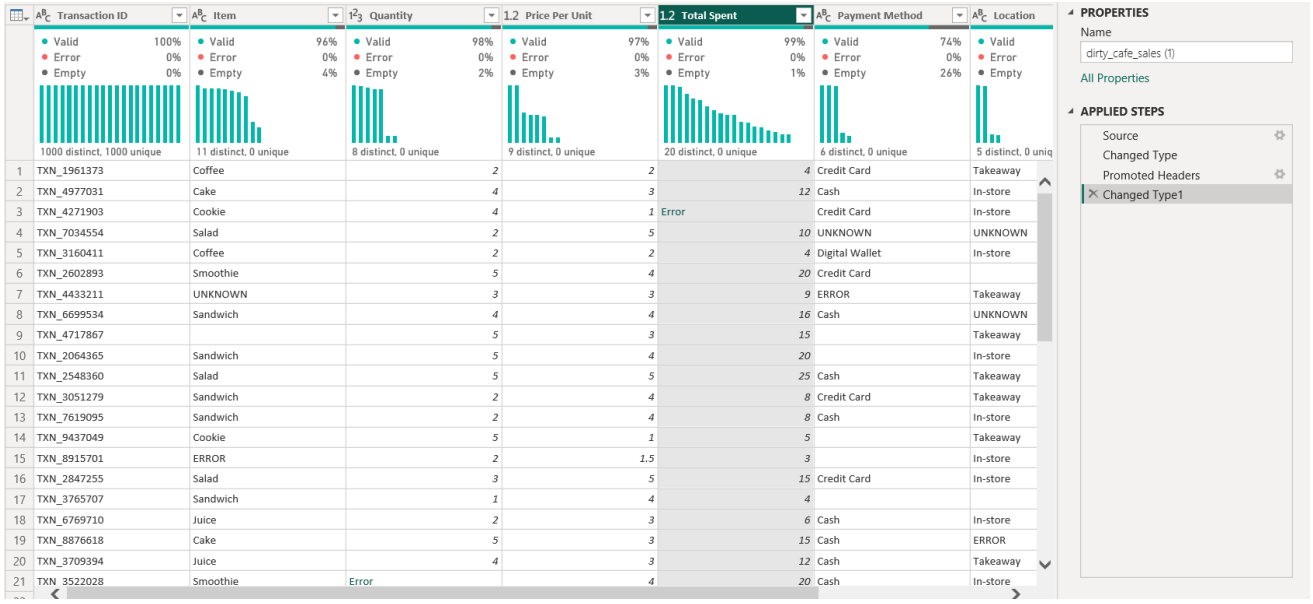
= Table.TransformColumnTypes(Source,{{"Column1", type text}, {"Column2", type text}, {"Column3", type text}, {"Column4", type text},

	Column1	Column2	Column3	Column4	Column5	Column6	Column7
	Valid 100%, Error 0%, Empty 0%, 1000 distinct, 1000 unique	Valid 96%, Error 0%, Empty 4%, 12 distinct, 1 unique	Valid 98%, Error 0%, Empty 2%, 9 distinct, 1 unique	Valid 97%, Error 0%, Empty 3%, 10 distinct, 1 unique	Valid 99%, Error 0%, Empty 1%, 21 distinct, 1 unique	Valid 74%, Error 0%, Empty 26%, 7 distinct, 1 unique	Valid, Error, Empty, 6 distinct, 1 uniq
1	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
2	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway
3	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store
4	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store
5	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN
6	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store
7	TXN_2602893	Smoothie	5	4.0	20.0	Credit Card	
8	TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway
9	TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN
10	TXN_4717867		5	3.0	15.0		Takeaway
11	TXN_2064365	Sandwich	5	4.0	20.0		In-store
12	TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway
13	TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway
14	TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store
15	TXN_9437049	Cookie	5	1.0	5.0		Takeaway
16	TXN_8915701	ERROR	2	1.5	3.0		In-store
17	TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store
18	TXN_3765707	Sandwich	1	4.0	4.0		
19	TXN_6769710	Juice	2	3.0	6.0	Cash	In-store
20	TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR
21	TXN_3709394	Juice	4	3.0	12.0	Cash	Takeaway

Figure 1. Initial Power Query data profiling showing valid, error and empty percentages.

4. Identified Data Problems

- Item contains blank and UNKNOWN/ERROR entries that should not be treated as genuine products.
- Quantity and Price Per Unit require numeric conversion and validation.
- Total Spent contains invalid entries in the raw data and should be checked against a calculated amount.
- Payment Method and Location contain missing/placeholder information.
- Transaction Date needs to be converted to a valid Date data type before time-based analysis.
- Transaction ID should be checked for uniqueness to prevent duplicate transactions.



	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location
	Valid 100% Error 0% Empty 0%	Valid 96% Error 0% Empty 4%	Valid 98% Error 0% Empty 2%	Valid 97% Error 0% Empty 3%	Valid 99% Error 0% Empty 1%	Valid 74% Error 0% Empty 26%	Valid Error Empty
	1000 distinct, 1000 unique	11 distinct, 0 unique	8 distinct, 0 unique	9 distinct, 0 unique	20 distinct, 0 unique	6 distinct, 0 unique	5 distinct, 0 uniq
1	TXN_1961373	Coffee	2	2	4	Credit Card	Takeaway
2	TXN_4977031	Cake	4	3	12	Cash	In-store
3	TXN_4271903	Cookie	4	1	Error	Credit Card	In-store
4	TXN_7034554	Salad	2	5	10	UNKNOWN	UNKNOWN
5	TXN_3160411	Coffee	2	2	4	Digital Wallet	In-store
6	TXN_2602893	Smoothie	5	4	20	Credit Card	
7	TXN_4433211	UNKNOWN	3	3	9	ERROR	Takeaway
8	TXN_6699534	Sandwich	4	4	16	Cash	UNKNOWN
9	TXN_4717867		5	3	15		Takeaway
10	TXN_2064365	Sandwich	5	4	20		In-store
11	TXN_2548360	Salad	5	5	25	Cash	Takeaway
12	TXN_3051279	Sandwich	2	4	8	Credit Card	Takeaway
13	TXN_7619095	Sandwich	2	4	8	Cash	In-store
14	TXN_9437049	Cookie	5	1	5		Takeaway
15	TXN_8915701	ERROR	2	1.5	3		In-store
16	TXN_2847255	Salad	3	5	15	Credit Card	In-store
17	TXN_3765707	Sandwich	1	4	4		
18	TXN_6769710	Juice	2	3	6	Cash	In-store
19	TXN_8876618	Cake	5	3	15	Cash	ERROR
20	TXN_3709394	Juice	4	3	12	Cash	Takeaway
21	TXN_3522028	Smoothie	Error	4	20	Cash	In-store

Figure 2. Cleaned Power Query table after data-type correction and error removal.

5. Creating the Calculated Total

After cleaning the numeric fields, a Custom Column was created in Power Query to independently calculate each transaction amount. This is an important validation step because it provides a second value against which the source Total Spent can be checked.

Calculated Total = [Quantity] × [Price Per Unit]

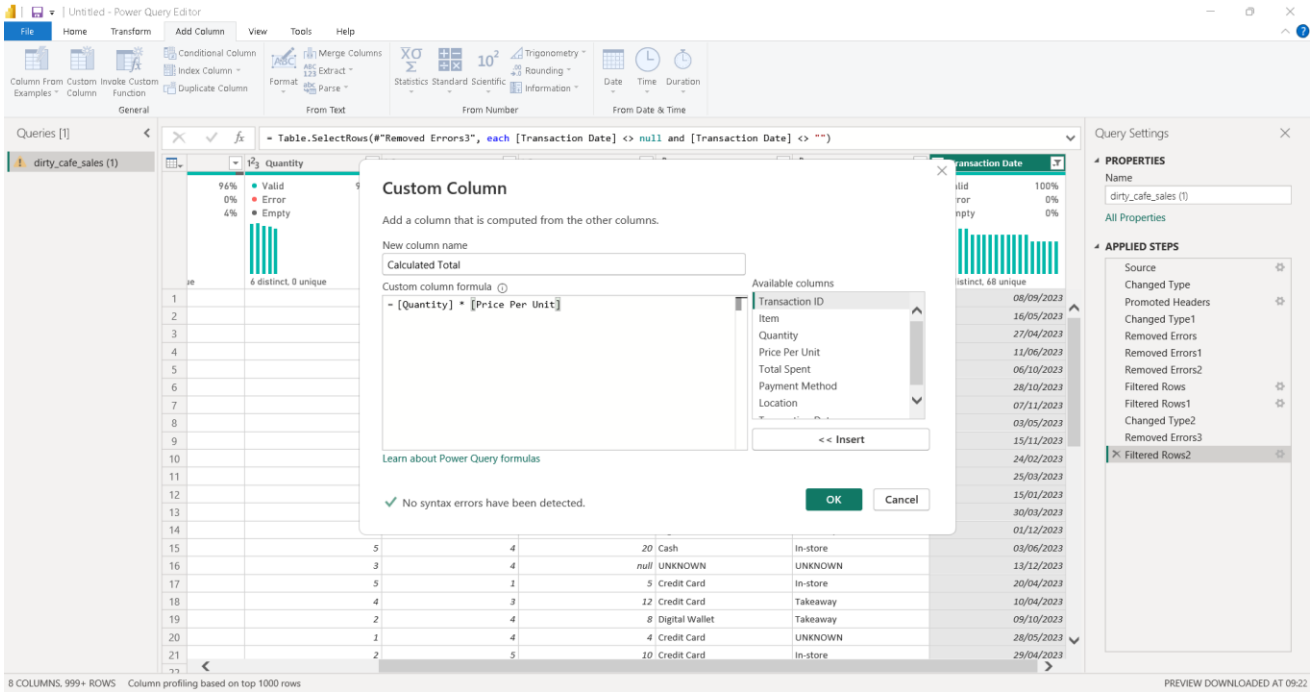


Figure 3. Power Query Custom Column formula used to calculate the transaction total.

6. Result of the Calculation

The calculated column was successfully created and converted to a numeric type. The preview shows that valid records produce the expected transaction amount—for example, a quantity of 4 at a unit price of 4 produces a calculated total of

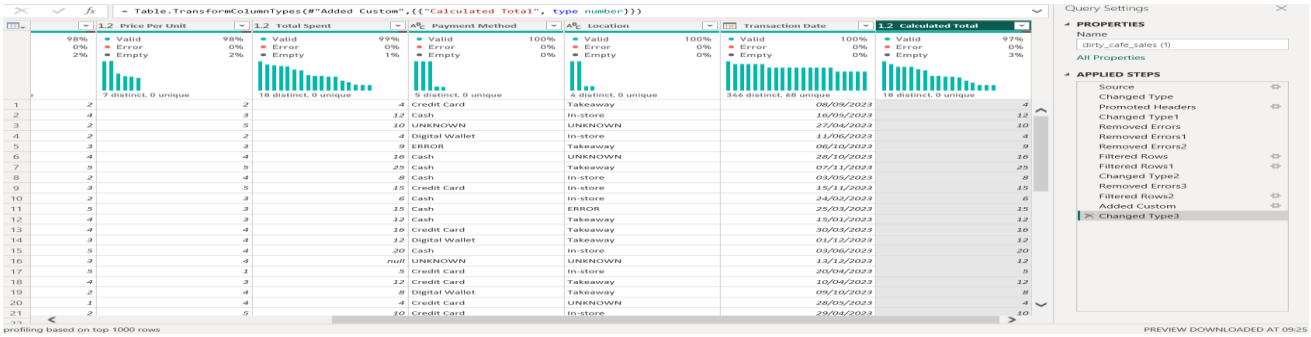


Figure 4. Calculated Total column after the custom calculation was applied.

7. Total Validation

A second Custom Column was created to compare Total Spent with Calculated Total. The validation rule classifies each row as Valid when the two values agree and Check when they do not. This makes the cleaning process auditable and allows problematic records to be identified before analysis.

if [Total Spent] = [Calculated Total] then "Valid" else "Check"

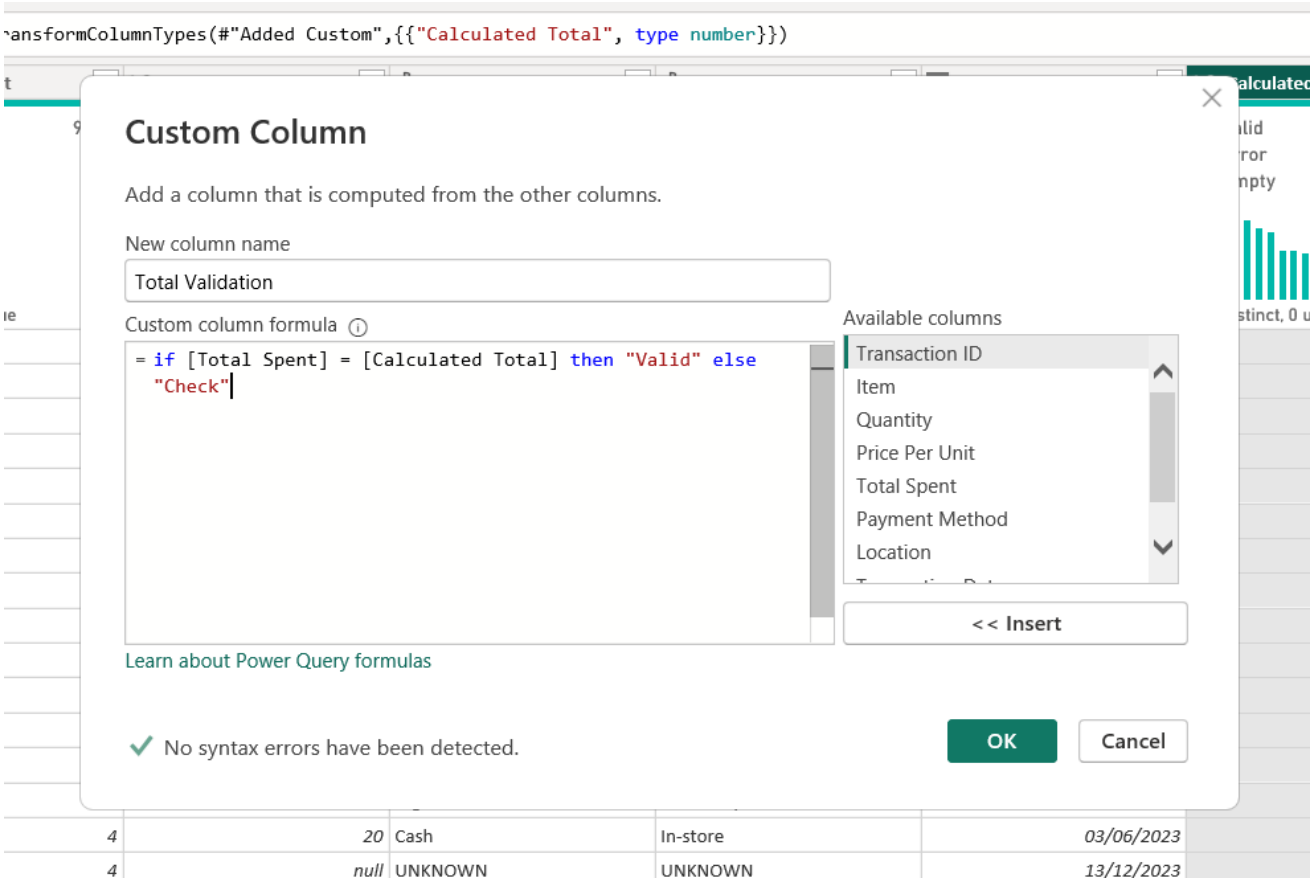


Figure 5. Custom Column formula used to compare Total Spent with Calculated Total.

8. Validation Results

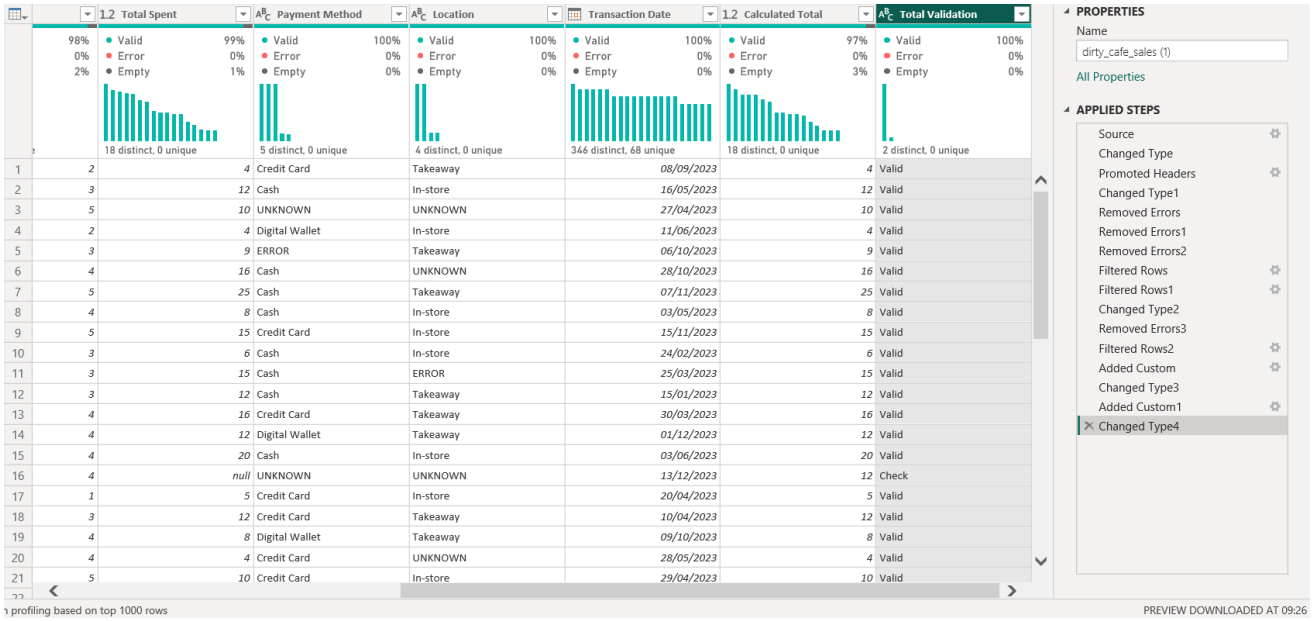


Figure 6. Final validation column showing Valid and Check classifications.

The screenshot demonstrates that most displayed rows are marked Valid. A Check result can occur when an input value was missing or could not produce a reliable calculated amount. Such records should be reviewed rather than silently included in financial totals.

9. Recommended Quality Rules

- Quantity must be a valid whole number greater than zero.
- Price Per Unit must be a valid positive numeric value.
- Calculated Total should equal Quantity multiplied by Price Per Unit.
- ERROR and UNKNOWN should be treated as invalid/missing values where appropriate.
- Transaction Date must be a valid date.
- Transaction IDs should be unique for transaction-level reporting.

10. Power BI Analysis and Dashboard Design

After the Power Query transformation is complete, the cleaned table can be loaded into the Power BI data model. The dashboard should focus on sales performance, transaction volume, products, payment methods, location and time.

11. Recommended Visuals

- KPI cards: Total Sales, Transaction Count, Units Sold and Average Transaction Value.
- Bar chart: Sales by Item to identify top-performing products.
- Line chart: Sales by Transaction Date to show trends over time.
- Bar/donut chart: Payment Method distribution using valid categories.
- Bar chart: Location comparison such as In-store versus Takeaway.
- Slicers: Date, Item, Payment Method and Location.

12. Conclusion

The Power Query workflow successfully demonstrates a practical data-cleaning and validation process. The raw transaction data was inspected, invalid records were handled, data types were corrected, a calculated transaction total was created, and a validation field was added to compare the calculated and source totals. This creates a stronger foundation for accurate Power BI reporting.

13. Final Recommendation

Before publishing the final dashboard, refresh the query, verify row counts and totals, review all Check records, and confirm that the visuals respond correctly to filters. Keeping the transformation steps in Power Query also makes the process repeatable when new transaction files are received.